



**Università
di Catania**

Amazon Prisma

Big Data Final Project

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Chapter 1

Introduction and Common Setup

1.1 Project Objective

1.1.1 Context

In the digital age, online reviews represent a source of invaluable data for companies and consumers. However, the volume of feedback generated daily is so vast that manual analysis is impossible. The goal of this project is to apply advanced *Natural Language Processing* (NLP) and unsupervised *machine learning* techniques to analyze a large corpus of Amazon product reviews. The purpose is to automatically extract the main themes discussed by users, group the reviews based on these themes (*clustering*), and analyze the sentiment associated with each group, thus providing strategic and actionable insights.

1.1.2 Dataset: Amazon Reviews

The dataset used is a version of the *Amazon Review Polarity Dataset*, originally derived from the *Stanford Network Analysis Project* (SNAP). The full collection contains approximately 36 million reviews gathered over an 18-year period, up to March 2013.

Attributes: Polarity, Title, Text

For each review, the dataset provides three fundamental attributes:

- **polarity:** A numerical label that indicates the review's sentiment. It can only take two values, 1 to indicate that the review is negative, and 2 to indicate that the review is positive.
- **title:** The title of the review, which often summarizes the user's judgment.
- **text:** The full body of the review, containing the details of the user's experience.

Sampling (Sample Size = 10,000)

Given the enormous size of the complete dataset, which would make an exhaustive analysis computationally impractical with currently available resources, a random sample of 10,000 records was extracted for this exploratory phase. This size is considered sufficient to test and validate the analysis pipelines while maintaining manageable execution times.

1.2 Development Environment and Prerequisites

1.2.1 Platform: Google Colab

The entire project was developed using Google Collaboratory (Colab). This platform was chosen for several key advantages:

- **Access to Free Resources:** It provides access to powerful CPUs and, especially, GPUs (*Graphics Processing Units*), which are fundamental for accelerating the calculations of modern NLP models based on Transformer architectures.
- **Pre-configured Environment:** Many of the necessary libraries are already installed, simplifying the setup phase.
- **Integration with Google Drive:** It allows for easy uploading and accessing of datasets stored in one's cloud space.

1.2.2 Installation of Main Libraries

The first phase of the code consists of installing all the necessary libraries via the pip package manager. The libraries can be grouped by function:

Data Manipulation: `pandas`, `numpy`

Natural Language Processing: `spacy` (with the backend for transformers models), `sentence-transformers`, `bertopic`

Machine Learning and Clustering: `scikit-learn` (for metrics and utilities), `umap-learn` (for dimensionality reduction), `hdbscan` (for clustering)

Data Visualization: `plotly` (for interactive graphs), `seaborn`, `matplotlib` (for static and custom graphs)

1.2.3 Authentication and Data Loading from Google Drive

To access the `train.csv.gz` file, the code authenticates to the user's Google account. Using the `google.colab` and `pydrive2` libraries, a secure connection to Google Drive is established, allowing the file to be downloaded directly to the Colab execution environment and loaded into a pandas DataFrame for analysis.

The dataset was made available in a public repository on Google Drive, accessible via a unique resource ID. This approach ensures immediate data access without the need for complex configurations, facilitating the reproducibility of the entire analysis process. Access via resource ID allows bypassing normal sharing restrictions, making the dataset easily accessible to all authorized users.

Chapter 2

Approach 1: Clustering Based on Named Entity Recognition (NER)

2.1 Logic of the Approach

The fundamental hypothesis of this first approach is that the central themes of a review are defined by the named entities (objects, products, brands, people, places) mentioned within it. Instead of analyzing the entire text, which can contain a lot of "noise" (common words, conjunctions, etc.), the focus is on the key elements that are the subject of the discussion. This feature engineering technique allows for reducing the complexity of the text and focusing the clustering algorithm on the most significant and discriminating concepts.

2.2 Entity Extraction and Preparation

2.2.1 Text Preprocessing

Before extracting entities, the raw text is cleaned and standardized through the `preprocess_ner` function. This function:

- Converts the text to ASCII format, ignoring non-standard characters.
- Removes all characters that are not alphanumeric or essential punctuation (like ., -, ').
- Normalizes multiple white spaces into a single space.

Subsequently, the `title` and `text` fields are combined into a single `combined_text_ner` field to have a complete context on which to perform the analysis.

2.2.2 Entity Extraction with spaCy

The core of this approach is entity extraction, performed with the spaCy library.

Model Used: `en_core_web_trf`

The `en_core_web_trf` model was chosen. Unlike lighter models, this one is based on a **Transformer** architecture, which gives it a greater ability to understand context and, consequently, higher accuracy in recognizing entities, especially in complex or informal texts like reviews. The code checks for GPU availability and sets it as the preferred device to accelerate model inference.

Filtering Relevant Entities (PRODUCT, ORG, etc.)

Not all entities are equally important. The `extract_entities` function is configured to extract only a subset of labels relevant to the context of Amazon reviews: `{'PERSON', 'ORG', 'GPE', 'PRODUCT', 'WORK_OF_ART', 'EVENT', 'LOC'}`

- **PRODUCT** and **WORK_OF_ART**: (for books, movies, etc.) are crucial.
- **ORG**: (organization) can identify manufacturers or brands.
- **PERSON**, **GPE** (geopolitical entity), **LOC** (location): can refer to authors, characters, or relevant geographical contexts.

Additionally, two subsequent filters are applied:

- **Cleaning**: Entities are converted to lowercase and initial/final punctuation is removed (`clean_entity`).
- **Exclusion**: Overly generic or common words (`common_words_to_ignore`) like "book", "movie", "product", "amazon" are removed, as they do not add semantic value for topic differentiation.

2.2.3 Creation of an Entity-only Corpus (`entity_text`)

The final result of this phase is a new column in the DataFrame, `entity_text`. Each row in this column contains a string composed solely of the filtered entities extracted from the corresponding review. This entity corpus will be the input for the clustering pipeline.

	combined_text_ner	entities
0	Expensive Junk. This product consists of a piece of thin flexible insulating material adhesive backed velcro and white electrical tape. Problems 1. Instructions are three pictures with little more information. 2. Velcro was all crumpled as received and was stronger than the adhesive. When I tried to disengage the velcro both pieces came off and the paint from the ceiling. 3. While electrical tape was horrible... cheap narrow and it fell off in less than 1 hour. 4. The price is a ripoff! I am building my own which is easier to use cheaper more attractive and higher r-value. I am surprised Amazon even lists this junk.	[]
1	Toast too dark. Even on the lowest setting the toast is too dark for my liking. Also the on light stays lit so you have to unplug it to avoid wasting electricity. Not the quality I expected from Cuisinart.	['cuisinart']
2	Excellent imagery...dumbed down story. I enjoyed this disc. The video is stunning. I agree with others that the story is very dumbed down and takes a childish approach. It actually seems like its a little one sided and VERY pro-environmental. Nevertheless its enjoyable. I would say however that the Amazon WMV HD disc is a better story and has better and sharper images and more interesting things to look at.	['wmv hd']

Figure 2.1: Example of filtered entities in the entities column.

2.3 Clustering Pipeline

Once the entity corpus is obtained, an unsupervised machine learning pipeline is applied to group the reviews.

2.3.1 Vectorization with TF-IDF

The entity text is transformed into a numerical matrix using `TfidfVectorizer`. TF-IDF (Term Frequency-Inverse Document Frequency) is a technique that assigns a weight to each entity. The weight is high if the entity is frequent in a single review (TF) but rare in the entire corpus of reviews (IDF). This allows for giving more importance to entities that are truly characteristic of a small group of documents.

2.3.2 Dimensionality Reduction with UMAP

The TF-IDF matrix is typically very large and sparse. To visualize it and to help the clustering algorithm, its dimensionality is reduced to 2 dimensions using **UMAP** (Uniform Manifold Approximation and Projection). UMAP is a powerful algorithm that excels at preserving both the local and global structure of the data, projecting the points into a low-dimensional space where similar reviews are close to each other.

2.3.3 Clustering with HDBSCAN

On the 2-dimensional embedding produced by UMAP, **HDBSCAN** (Hierarchical Density-Based Spatial Clustering of Applications with Noise) is applied. Its main advantages are:

- It does not require pre-specifying the number of clusters.
- It is density-based and can identify clusters of arbitrary shapes.
- It can mark points that do not belong to any dense cluster as noise (label -1), making the analysis more robust.

In the context of entity-based thematic clustering, we can note how the red cluster on the left highlights a phenomenon of high variance in the extracted entities. This cluster seems to contain reviews with random or weakly correlated entities that do not form a clear or coherent thematic structure. For example:

- Some reviews may mention a few relevant entities.

- Others may contain common and generic entities (e.g., "product", "amazon", "book"), which do not add semantic value for topic differentiation.
- Reviews with random or weakly correlated entities can be difficult to group into coherent themes, leading to a "diffuse" and unstructured cluster.

This red cluster therefore represents the noise in the dataset, where the reviews do not align with any of the main themes identified by the other clusters. HDBSCAN correctly identified this set of points as a less dense cluster, marking it as noise.

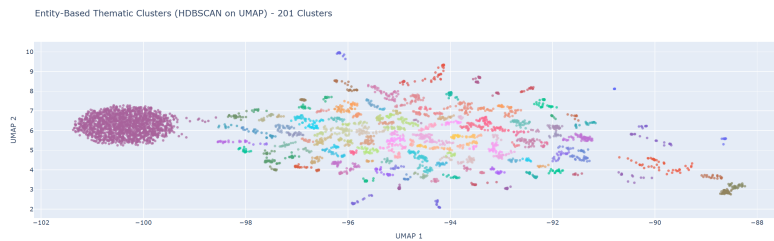


Figure 2.2: Entity-based Thematic Clusters.

2.4 Analysis and Visualization of Results

2.4.1 Cluster Evaluation Metrics

To quantitatively evaluate the quality of the clustering (excluding noise points), three standard metrics are calculated:

- **Silhouette Score:** Measures how similar an object is to its own cluster compared to other clusters. A high value indicates good separation.
- **Calinski-Harabasz Index:** Based on the ratio of between-cluster dispersion to within-cluster dispersion. The higher, the better.
- **Davies-Bouldin Index:** Measures the average similarity of each cluster to its most similar cluster. The lower, the better.

2.4.2 Interactive Cluster Visualization with Plotly

The clustering result is visualized via an interactive scatter plot created with `plotly.express`. Each point on the graph represents a review, positioned

according to its UMAP coordinates and colored based on the cluster label assigned by HDBSCAN. By hovering over a point, you can view the original review text and the extracted entities, facilitating the exploration and qualitative interpretation of the clusters.

2.4.3 Sentiment and Entity Analysis per Cluster

For each identified cluster, a summary table (`cluster_summary`) is created that shows:

- **num_reviews**: The size of the cluster.
- **average_sentiment**: The average polarity (normalized to 0-1), which indicates whether the cluster's theme is predominantly discussed in positive or negative terms.
- **principal_entities**: The 10 most representative entities of the cluster, extracted via TF-IDF.

cluster_label	num_reviews	average_sentiment	principal_entities
11	11	1720	0.518023 blade, cannon, chii, dexter, donny, oil, oracle, pampers, quicken, uniden
31	31	116	0.517241 clooney, hd, oregon, roc, soc, stephen, stitch, sunforce, teacher, todd
72	72	60	0.500000 kt, matthew, night, queen, reserve, secrets, silence, street, tunstall, world
172	172	56	0.607143 bbc, clark, good, great, jimmy, life, pat, security, simon, van
77	77	47	0.340426 360, america, england, junk, larry, london, microsoft, vista, windows, xbox
83	83	45	0.511111 bea, chris, ii, india, java, john, kindle, limbaugh, otis, rush
81	81	38	0.605263 america, bellesiles, che, chef, columbia, grisham, hamburger, john, south, university
136	136	38	0.447368 angeles, bay, hard, jam, los, pearl, sarah, street, titans, wonder
162	162	38	0.605263 bridges, dickens, dunham, enigma, francisco, hardy, jeff, john, mars, san
103	103	37	0.405405 better, body, book, bureau, business, charmed, cook, hal, shadows, twilight
125	125	37	0.729730 art, beauty, bose, buns, buzzy, electronica, endymion, fan, freedom, hyperion
2	2	37	0.405405 32hs420, a100, b1000, betta, dg810, dsir, laurmea, mdr, s560d, sony
128	128	33	0.636364 babe, boss, crue, devil, dirt, east, great, heaven, motley, priest
90	90	32	0.593750 australia, death, dogg, flesh, harvey, kiss, mitchell, norther, pound, zen
148	148	32	0.593750 alain, delon, finding, focusfilm, green, holmes, russell, sherlock, vivian, watson
158	158	32	0.687500 angels, brown, clifford, dan, demons, james, jill, jim, roach, thomas
107	107	31	0.483871 adam, arthur, christmas, crimson, elmo, king, lothar, sandler, stephen, swan
134	134	31	0.612903 alice, aristotle, chains, evangelion, fiction, germany, girl, redd, reich, wonderland
156	156	31	0.677419 eric, jellyfish, johnson, kenny, martha, messengers, nick, paul, soards, wilson
109	109	30	0.533333 ballard, dennis, edith, engling, gentle, google, kate, magdalena, mary, parker

Figure 2.3: Sentiment and Entity Analysis per Cluster.

2.4.4 Analysis of Entity Co-occurrence (Pairs and Triplets)

Finally, a more granular analysis is conducted to discover relationships between entities. The code identifies all pairs and triplets of entities that appear

together in multiple reviews and, for each combination, calculates the number of occurrences and the associated average *sentiment*. This type of analysis can reveal specific *insights*, such as whether the combination “iPhone” and “battery” is associated with a more negative sentiment compared to the combination “iPhone” and “camera”.

	Entità_1	Entità_2	Entità_3	Conteggio	Sentiment_Medio
1238	a walk to remember	sparks	the notebook	2	1.0
20425	abbott	costello	universal	2	2.0
25179	courtney cox	neve campbell	scream	2	1.0
5748	message in a bottle	sparks	the notebook	2	1.0
11888	bush	fahrenheit 911	moore	2	1.5

Figure 2.4: Triplets found in the subset under examination.

Colab Notebook available at the following link:

<https://colab.research.google.com/drive/1Y0W8jPOHq-c2LII7WwTq7jYEfcJGfKRw?usp=sharing>

Chapter 3

Approach 2: Semantic Analysis with Sentence-Transformers

This chapter explores an alternative approach based on latent semantic analysis, leveraging *Sentence-Transformers* models. Unlike entity-based analysis, which focuses on explicit names and concepts, this method aims to capture the full contextual meaning of a text, allowing documents to be grouped based on their conceptual similarity rather than simple keyword co-occurrence.

3.1 Logic of the Approach

The fundamental idea guiding this approach is that the meaning of a review resides not just in the mentioned entities (e.g., "battery", "screen"), but in the entire semantic context.

The hypothesis is that themes can be discovered by analyzing the complete semantic meaning of the text, not just the explicit entities.

To test this hypothesis, each document is transformed into a high-dimensional numerical vector, known as an **embedding**. These vectors, generated by a pre-trained model, capture the semantic nuances of the text. Documents with similar meanings will have vectors that are close in the vector space, regardless of the specific lexicon used.

3.2 Creating Semantic Embeddings

The first step is to transform the textual corpus into a numerical format that can be processed by machine learning algorithms. This process is divided into two phases: a preliminary text cleaning and its subsequent vector encoding.

3.2.1 Text Preprocessing

Before embeddings can be generated, the raw text is subjected to a cleaning phase to standardize the input. This logic is implemented in a Python function, called `preprocess_text`, which is applied to both the title and the text of each review before combining them into a single text field.

3.2.2 Text Encoding with Sentence-Transformers

Once the text is cleaned, it is encoded into numerical vectors.

Model Used: `all-MiniLM-L6-v2`

The `all-MiniLM-L6-v2` model was chosen for generating the embeddings. Using the `Sentence-Transformers` library, the model is instantiated as an `sbert_model` object, taking care to use GPU acceleration (`'cuda'`) if available. This model maps sentences and paragraphs into a 384-dimensional vector space.

Generating Embedding Vectors

The loaded model is then used to encode the entire corpus. By calling the `sbert_model.encode()` method on the list of texts, a NumPy matrix, `full_text_embeddings`, is obtained, where each row represents the vector of the corresponding document.

3.3 Topic Modeling and Thematic Analysis

With the embeddings available, groups of thematically similar reviews are identified through a clustering pipeline.

3.3.1 Dimensionality Reduction (UMAP) and Clustering (HDBSCAN) on Embeddings

The embeddings are first projected into 2D with **UMAP** and then grouped with **HDBSCAN**. From an implementation perspective, a `umap.UMAP` reducer is initialized with parameters such as `n_neighbors=40` and `metric='cosine'`, and subsequently an `hdbscan.HDBSCAN` object with `min_cluster_size=15`.

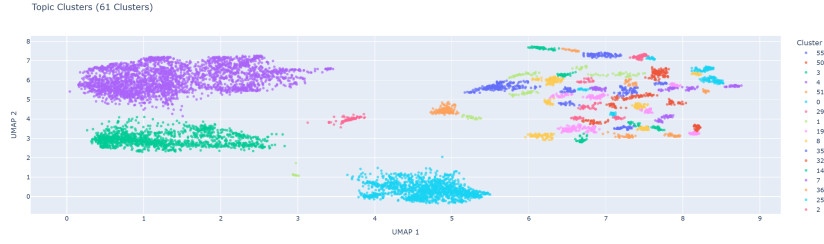


Figure 3.1: Thematic clusters found with Sentence-Transformer.

3.3.2 Theme Extraction with c-TF-IDF

To understand the content of each cluster, **c-TF-IDF** is used. This method, implemented through a custom function, extracts the most representative terms for each theme. The analysis made it possible to identify very specific and coherent thematic clusters. For example:

- **Cluster 8:** Focused on *"dvd, player, play, discs, cd"*. This cluster was found to have an extremely negative average sentiment (0.11 out of 1), indicating that it mainly groups complaints about faulty DVD players and discs.
- **Cluster 34:** Whose main themes are *"coffee, water, maker, carafe, pot, cup"*. This group, with an average sentiment of 0.32, gathers discussions about coffee machines, highlighting both problems and functional aspects.
- **Cluster 10:** Related to *"printer, print, ink, cartridge, hp"*. This cluster also has a predominantly negative sentiment (0.25), suggesting that the main topic is problems related to printers and consumables.

These results demonstrate the effectiveness of the method in autonomously grouping reviews not only by product but also by problem type.

cluster_label	num_reviews	avg_sentiment	main_topics	
4	4	3000	0.533333	book, read, books, reading, story, author, written, good, characters, like
0	0	1493	0.634293	album, cd, music, songs, song, like, band, sound, best, great
3	3	1410	0.458156	movie, film, dvd, movies, watch, acting, great, just, good, funny
35	35	208	0.552885	toy, old, daughter, son, seat, fun, toys, baby, loves, kids
6	6	180	0.550000	game, games, graphics, play, fun, gameplay, playing, played, like, really
50	50	124	0.435484	pan, oven, use, toaster, used, nonstick, stainless, pans, handle, dishwasher
19	19	114	0.298246	drive, product, computer, card, software, mac, windows, files, usb, gb
12	12	103	0.563107	taste, product, tea, vitamins, vitamin, organic, soup, taking, food, flavor
2	2	103	0.631068	workout, yoga, dvd, pilates, video, exercises, exercise, moves, body, dance
49	49	102	0.441176	jeans, fit, diapers, bed, soft, pad, size, sheets, pants, material
25	25	93	0.462366	vacuum, brush, filter, water, filters, use, cleaner, cleaning, mop, floor
34	34	84	0.321429	coffee, water, maker, carafe, pot, cup, plastic, machine, grind, brew
14	14	83	0.662651	recipes, food, cookbook, cooking, recipe, cook, book, cookbooks, fat, dishes
8	8	78	0.115385	dvd, player, play, discs, disc, cd, software, dvds, bought, program
17	17	71	0.535211	skin, scent, smell, product, perfume, oily, bottle, oil, smells, love
55	55	67	0.432836	product, cheap, shower, door, broke, machine, price, rope, use, extender
30	30	64	0.515625	shipping, item, product, order, arrived, fast, shipped, received, thanks, ordered
10	10	63	0.253968	printer, print, ink, cartridge, hp, paper, cartridges, printers, epson, toner
54	54	59	0.322034	lock, gun, key, item, product, ammo, pin, keys, received, came
22	22	54	0.259259	phone, headset, phones, handsets, static, nokia, handset, base, cell, sound
23	23	52	0.576923	headphones, sound, ear, pair, ears, quality, buds, noise, earphones, headset
40	40	52	0.326923	light, bulb, lamp, lights, bright, amazon, product, poor, switch, hole
47	47	45	0.422222	cable, cables, hdmi, tv, product, receiver, works, signal, work, audio

Figure 3.2: Examples of themes extracted from the subset with c-TF-IDF.

3.4 Applications and Advanced Analyses

This methodology allows for several high-level analyses.

3.4.1 Implementing a Semantic Search System

The embeddings form the basis for a search system that finds relevant documents based on meaning. The logic is encapsulated in a `find_similar_reviews` function that uses cosine similarity. The effectiveness of this approach is evident in the results: a search for the concept of “buon rapporto qualità-prezzo” (good value for money) is not limited to searching for these exact words. The results returned include reviews with phrases like *"value for money, definately!"* and *"perfect for the price"*, demonstrating the system’s ability to understand the semantic intent of the query.

Results for a concept (value for money):				
	similarity_score	combined_text	cluster_label	sentiment_label
6665	0.575079	value for money, definately! an excellent camcorder!! i have nothing bad to say about its performance, although an lcd screen would have been convenient.	44	Positive
4018	0.504719	perfect for the price.. i bought them because you cannot beat the price. i figured i could get a minimum of a year out them. have used them a couple of times and i am impressed, sure they are chea...	54	Positive

Figure 3.3: Example result for the concept (value for money).

3.4.2 Analyzing Semantic Coherence between Title and Text

The semantic coherence between the title and body of each review was measured by calculating the cosine similarity between their embeddings. The analysis revealed a distribution of scores (shown in Figure 4.1) with a peak around low similarity values (0.1-0.2), suggesting that, on average, review titles are not very faithful semantic summaries of their content.

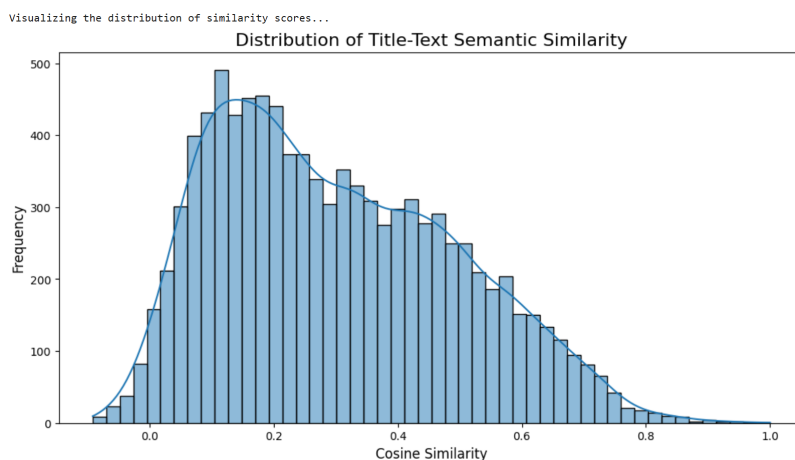


Figure 3.4: Histogram of the distribution of cosine similarity scores between review title and text.

A manual inspection of the distribution's extremes proved to be particularly insightful. Reviews with **very high coherence** (scores > 0.9) are often those where the title is an exact copy or a literal paraphrase of the text. Conversely, reviews with **very low coherence** (scores < 0) often have ironic, misleading, or titles that express a marginal concept compared to the body of the review. This phenomenon highlights the complexity and sometimes unreliability of titles alone as indicators of the actual content.

3.4.3 Detailed Inspection of Individual Clusters

Finally, for a qualitative analysis, a support function, `inspect_cluster`, was created. By providing this function with a cluster ID, it retrieves key information and prints examples of reviews, facilitating the interpretation and validation of the emerging themes.

Colab Notebook available at the following link:

<https://colab.research.google.com/drive/16Q1QvTVTxrrnSr1LS-M7JciLQx2QQXHZ?usp=sharing>

Chapter 4

Approach 3: Hybrid Model with BERTopic and Text Enrichment

This approach implements an advanced Natural Language Processing (NLP) pipeline to analyze the Amazon product review dataset. The goal is to extract significant discussion topics and analyze their associated sentiment. The approach stands out for a preliminary text enrichment phase, designed to improve the quality of subsequent topic modeling.

4.1 Logic of the Approach

The underlying hypothesis is that semantic topic modeling models like BERTopic, while powerful, can benefit from more explicit input text. Enriching the original text with named and contextualized entities (e.g., transforming "Apple" into "Apple company" or "iPhone" into "iPhone product") helps the model capture nuances that might otherwise be lost, leading to the generation of clearer, more robust, and more detailed topics.

4.2 Hybrid Text Enrichment Pipeline

The core of this script's innovation lies in its enrichment pipeline, which combines two complementary techniques: Named Entity Recognition (NER) and Entity Linking.

4.2.1 Entity Extraction with Transformer-based NER

The first step is to identify named entities (people, organizations, products, locations) present in the review text.

Model Used: `dbmdz/bert-large-cased-finetuned-conll103-english`

This script uses a powerful model based on the BERT architecture, pre-trained and then fine-tuned on the CoNLL-2003 dataset for the NER task in

English. It is loaded via the `transformers` library. The `extract_entities_ner_with_positions` function not only extracts entities but also identifies their exact position in the text, filtering the most relevant ones (such as ORG, PRODUCT, LOC) and discarding noise or overly common words.

4.2.2 Entity Linking with TagMe API

While NER identifies an entity, Entity Linking connects it to a unique entry in a knowledge base (like Wikipedia), resolving ambiguities.

Contextualization and Disambiguation of Entities

The script uses the TagMe service for this purpose.

The `extract_tagme_with_positions` function sends the text to the TagMe API, which returns the mentioned entities linked to the corresponding Wikipedia pages. This process is crucial for disambiguating generic terms (e.g., "Apple" could refer to the fruit or the company; TagMe contextualizes it correctly). The code filters TagMe annotations with a confidence score ($\rho > 0.15$) to ensure quality.

4.2.3 Merging and Contextual Enrichment

The `enrich_text_contextually` function is the point of union. It takes entities from NER and TagMe and inserts them into the original text intelligently:

- **Manages overlaps**, giving priority to TagMe, as it provides a more specific enrichment (e.g., a Wikipedia link is more informative than a generic "ORG" label).
- **Adds context**:
 - For NER entities, it adds a qualifier (e.g., iPhone → iPhone product).
 - For TagMe entities, it expands the mention with the Wikipedia page title (e.g., Jobs → Jobs Steve Jobs).
- **Cleans the final text** by removing artifacts and parentheses to prepare it for the next phase.

4.3 Topic Modeling with BERTopic

Once the enriched text corpus is obtained, it is given as input to BERTopic for topic extraction.

4.3.1 The Concept of BERTopic as an Integrated Framework

BERTopic is not a single algorithm, but a modular framework that combines different techniques:

- **Text Embedding:** It uses **sentence-transformers** (specifically, the **all-MiniLM-L6-v2** model) to convert each enriched review into a numerical vector that captures its semantic meaning.
- **Dimensionality Reduction:** It applies the **UMAP** (Uniform Manifold Approximation and Projection) algorithm to reduce the dimensionality of the embeddings, making clustering easier. Although it is also possible to use **PCA** (Principal Component Analysis) in BERTopic as an alternative technique, UMAP was preferred as it is more effective at preserving both the local and global structure of high-dimensional data, which is particularly advantageous in the case of complex and non-linearly distributed textual datasets, such as the one analyzed in this study.
- **Clustering:** It groups the vectors using the **HDBSCAN** algorithm, which can identify clusters of varying density and handle noise (outliers). Each cluster represents a topic.
- **Topic Representation:** It extracts the most representative keywords for each topic using a strategy based on **c-TF-IDF** (Class-based TF-IDF), which highlights terms important for one cluster compared to all others.

4.3.2 Objective: Providing Enriched Text to BERTopic

The entire enrichment process is aimed at this step. Instead of training BERTopic on raw text, it is trained on semantically denser and less ambiguous texts. This helps the model create more coherent and separate clusters.

4.3.3 Expected Result: More Robust and Detailed Topics

The final output is a list of topics. Thanks to the enrichment, it is expected that these topics will be:

- **Less generic:** For example, instead of a generic topic on "companies", distinct topics for "technology companies" and "publishing houses" could be obtained.
- **Easier to interpret:** The keywords of each topic are more specific.
- **More stable:** Enrichment reduces noise, leading to more stable and significant clusters.

General Topic Information:			Representation
Topic	Count	Name	
0	0	0_book_read_reading_books	[book, read, reading, books, story, author, written, characters, life, writing]
1	1	1_album_cd_music_songs	[album, cd, music, songs, company, song, sound, rock, recording, record]
2	2	2_movie_film_dvd_movies	[movie, film, dvd, movies, watch, acting, series, funny, seen, story]
3	3	3_game_games_video_games_graphics	[game, games, video game, graphics, game game, video, play, fun, playing, playstation]
4	4	4_coffee_water_cup_maker	[coffee, water, cup, maker, coffee maker, blender, mug, cups, hot, pot]
5	5	5_toy_old_fun_daughter	[toy, old, fun, daughter, year old, toys, son, kids, year, loves]
6	6	6_workout_exercise_yoga_dvd	[workout, exercise, yoga, dvd, pilates, physical exercise, physical, exercises, video, moves]
7	7	7_recipes_food_cookbook_book	[recipes, food, cookbook, book, cooking, recipe, bread, cook, cookbooks, dishes]
8	8	8_tool_knife_saw	[tool, blade, knife, saw, cut, cutting, mower, handle, blades, knives]
9	9	9_dvd_player_disc_dvd_player	[dvd, player, disc, dvd player, play, cd, discs, company, software, unit]
10	10	10_vacuum_cleaner_brush_clean	[vacuum, cleaner, brush, clean, vacuum cleaner, cleaning, carpet, suction, mop, floor]
11	11	11_baby_bed_mattress_pad	[baby, bed, mattress, pad, sheet, diapers, sheet, diaper, potty, air]
12	12	12_worried_don_junk_unit	[worried, don, junk, unit, piece, sent, days, broke, months, stopped]
13	13	13_printer_cartridge_ink_printing	[printer, cartridge, ink, printing, print, Epson, hp, paper, cartridges, Epson, velco]
14	14	14_pan_cookware_cookware_bakeware_bakeware	[pan, cookware, cookware bakeware, bakeware, oven, stainless, non stick, cake, pans, grill]
15	15	15_cheap_broke_metal_plastic	[cheap, broke, metal, plastic, junk, door, tape, heavy, metal, piece]
16	16	16_headphones_ear_sound_pair	[headphones, ear, sound, pair, noise, carry, head, hearing, headset, loss]
17	17	17_jeans_shirt_fit_wear	[jeans, shirt, fit, wear, pants, material, size, pair, fashion, waist]
18	18	18_shoes_wear_boots_shoe	[shoes, wear, boots, shoe, feet, pair, comfortable, rubber, socks, size]
19	19	19_drive_computer_hard_disk	[drive, computer, hard, disk, usb, disk drive, hard disk, software, drive hard, hard drive]
20	20	20_bag_backpack_bag_straps	[bag, backpack, bags, straps, strap, rack, space, zippers, bag bag, holdo]
21	21	21_cable_cables_hDMI_tv	[cable, cables, HDMI, tv, television, connector, audio, sony, connect, cable cable]
22	22	22_fast_shipped_freight_transport_freight	[fast, shipped, freight transport, freight, transport, arrive, ordered, com, package, happy]
23	23	23_watch_wrist_strap_bраслет	[watch, wrist, strap, bracelet, time, looks, band, date, clock, light]
24	24	24_scent_smell_perfume_bottle	[scent, smell, perfume, bottle, cologne, fragrance, smells, shower, bottles, oil]
25	25	25_taste_organic_flavor_chocolate	[taste, organic, flavor, chocolate, soup, food, tastes, miso, eating, haru]
26	26	26_camera_pictures_kodak_picture	[camera, pictures, kodak, picture, battery, digital, canon, cameras, digital camera, easy]
27	27	27_shave_razor-trimmer-shaver	[shave, razor, trimmer, shaver, close shave, close, beard, plug, robot, norelco]
28	28	28_light_bulb_lamp_light_bulb	[light, bulb, lamp, light bulb, lights, shade, bright, incandescent, incandescent light, solar]
29	29	29_christmas_bastles_movie_santa	[christmas, bastles, movie, santa, christmas medium, dvd, holiday, mr bean, bean, medium]
30	30	30_gps_global_global_positioning_positioning	[gps, gps global, global positioning, positioning, global, garmin, mount, naps, garmin company, navigation]
31	31	31_router_linksys_wireless_linksys_company	[router, linksys, wireless, linksys company, network, support, tech, computer, firmware, netgear company]
32	32	32_player_mp3_ipo_ipod_player	[player, mp3, ipod, ipod player, zune, apple, rio, cd, portable, device]
33	33	33_dog_toy_puppy_dogs	[dog, toy, puppy, dogs, harness, treats, lab, loves, collar, toys]
34	34	34_cats_cat_litter_dog	[cats, cat, litter, dog, camel, house, trap, fly, dogs, plastic]
35	35	35_vitamin_vitamin_gills_tasting	[vitamin, vitamin, gills, tasting, ash, food, tried, tests, follicle acid, follicle]
36	36	36_hair_conditioner_iron_dryer	[hair, conditioner, iron, dryer, hair dryer, scalp, shampoo, curling, flat iron, curl]
37	37	37_speakers_sound_speaker_base	[speakers, sound, speaker, stereo, base, bassline, bassline, speakers loudspeaker, stands]
38	38	38_skin_acne_oily_products	[skin, acne, oily, products, face, results, dermatitis, emoi, sensitive, dose]
39	39	39_battery_batteries_charge_canon	[battery, batteries, charge, canon, charger, rechargeable, company battery, unit, bnu, battery life]
40	40	40_phone_telephone_mobile_cellless	[phone, telephone, mobile, cordless, phone, handset, battery, mobile phone, mobile telephone]
41	41	41_picture_color_looks_ring	[picture, color, looks, ring, mask, pendant, poster, frame, swarovski, necklace]

Figure 4.1: BERTopic on our subset.

4.4 Advanced Analysis and Visualization of Results

In addition to the main workflow, the code performs several secondary analyses to explore the results in depth.

- **Polarity Analysis:** It cross-references the extracted topics with the original polarity of the reviews (positive/negative). This allows for the calculation of a "Net Sentiment Score" for each topic, quickly identifying which topics generate more complaints (e.g., "shipping and returns") and which generate more satisfaction (e.g., "audio quality"). The results are shown in both tabular and graphical form.

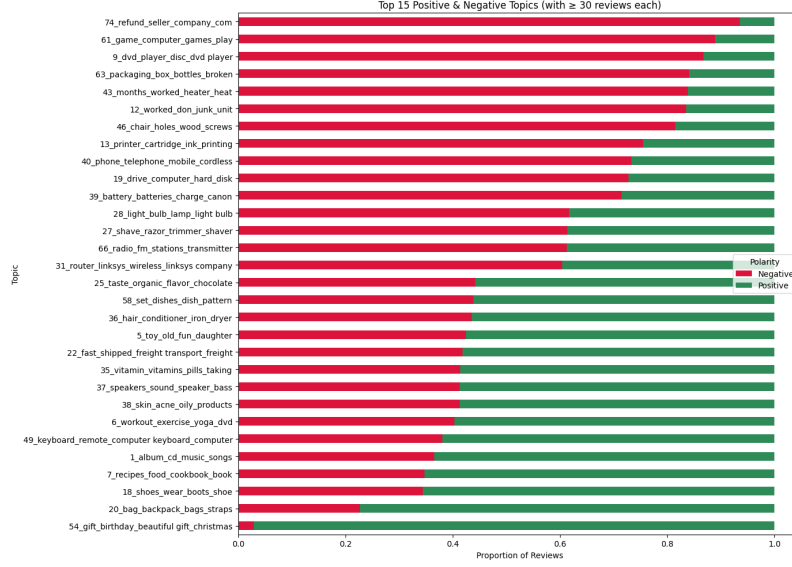


Figure 4.2: Top 10 Positive and Negative Topics from the subset.

- **Emotion Analysis:** It uses a text classification model (`j-hartmann/emotion-english-distilroberta-base`) to associate a dominant emotion (joy, anger, sadness, etc.) with each review. By aggregating this data by topic, it is possible to understand the emotional nuances related to each topic.
- **Aspect-Based Sentiment Analysis (ABSA):** It implements the most granular analysis. Using a zero-shot classification model (`facebook/bart-large-mnli`), it analyzes reviews to identify the sentiment associated with specific aspects of a product (e.g., price, battery, design, quality). This makes it possible to discover, for example, that within a topic on "smartphones", the battery is viewed negatively while the screen is viewed positively.
- **Interactive Visualizations:** It leverages BERTopic's powerful visualization functions to generate interactive graphs that show the relationship between topics, their hierarchy, the most important keywords, and the distribution of documents in the semantic space.

4.5 Conclusions and Analysis Results

The application of the hybrid pipeline on a dataset of 10,000 reviews allowed for the quantitative validation of the effectiveness of the proposed approach.

The preliminary text enrichment, combined with BERTopic and subsequent sentiment analyses, produced detailed and operational insights. The conclusions that emerged from the various phases of the analysis are reported below.

4.5.1 Interpretation of Polarity Results

The polarity analysis, quantified by the **Net Sentiment Score**, unequivocally revealed the areas of greatest customer criticality and satisfaction.

- **Topics with Strong Negative Sentiment:** The results show that the most significant issues are often related to defective products, damage during shipping, or negative purchasing experiences. For example, **Topic 74** (`refund_seller_company_com`) has the highest proportion of negative reviews, indicating major issues with refunds and the relationship with sellers and companies. Malfunctions of electronic products are also a significant cause of dissatisfaction, as confirmed by **Topic 9** (`dvd_player_disc_dvd_player`) and **Topic 43** (`months_worked_heater_heat`), associated with devices that stop working. Finally, the problem of products arriving damaged is clearly highlighted by **Topic 63** (`packaging_box_bottles_broken`).
- **Topics with Strong Positive Sentiment:** Conversely, the topics with the most positive sentiment are predominantly associated with gifts, hobbies, and personal items. **Topic 54** (`gift_birthday_beautiful_gift_christmas`) stands out above all, with an almost entirely positive proportion of reviews, demonstrating the great success of products purchased as gifts. Categories related to hobbies, such as cooking (**Topic 7**, `recipes_food_cookbook_book`) and music (**Topic 1**, `album_cd_music_songs`), also show a very high level of satisfaction. Finally, there is also strong appreciation for personal items like footwear and accessories, highlighted by the favorable feedback for **Topic 18** (`shoes_wear_boots_shoe`) and **Topic 20** (`bag_backpack_bags_straps`).

4.5.2 Emotional Analysis: Beyond Simple Sentiment

The emotion analysis provided a more nuanced understanding of user reactions, going beyond the simple positive/negative dichotomy.

- In the **most negative topics**, such as **Topic 74** (refund_seller_company_com) and **Topic 9** (dvd_player_disc_dvd player), a dominant emotion is **sadness**. In this context, this can be interpreted as disappointment with a faulty product or a negative customer service experience. This provides a more nuanced insight than a simple "negative" sentiment.
- In the **most positive topics**, particularly **Topic 54** (gift_birthday_beautiful gift_christmas), the predominant emotion is overwhelmingly **joy**. This clearly confirms the enthusiasm and satisfaction of users when purchasing products as gifts.

This analysis confirms that the approach is capable of capturing emotional nuances, providing a more complete psychological picture of the customer experience.

4.5.3 Insights with Aspect-Based Sentiment Analysis (ABSA)

Aspect-Based Sentiment Analysis (ABSA) has proven to be an extremely effective tool for obtaining high-resolution insights within the identified topics. Thanks to this approach, it was possible not only to understand the sentiment associated with a general topic but also to distinguish how each specific aspect (e.g., price, quality, sound, shipping) is perceived by users.

In particular, the analysis highlighted how some aspects are recurring sources of criticism. For example, in **Topic 2**, the aspects sound (78 negative reviews versus 40 positive) and price (69 negative versus 31 positive) show a clear prevalence of negative sentiment. Conversely, aspects like design and quality receive more varied evaluations: in **Topic 0** they are balanced, while in **Topic 1** they are predominantly positive (60 positive versus 42 negative for design and 74 versus 52 for quality, respectively).

This granularity is crucial for businesses: it allows them to take targeted action on the features that most influence the perception of the product or service, thus optimizing the user experience and commercial strategy.

Colab Notebook available at the following link:

<https://colab.research.google.com/drive/11NsBa6ik4ipRdJXQl0xRjNWm2P0o1pmo?usp=sharing>

Chapter 5

Conclusions and Future Developments

The analysis conducted on a subset of 10,000 reviews yielded significant results, despite the relatively small size of the sample. This choice represented a compromise between the need for statistically relevant data and the limitations imposed by available computational power.

Nevertheless, the investigation generated interesting insights and highlighted the validity of the methodological approach adopted, suggesting the presence of useful and coherent patterns within the dataset. The results obtained on this subset indicate that the approaches are promising and can be further refined and enhanced.

In terms of future developments, it would be appropriate to extend the analysis to a larger dataset. An increase in the volume of data under examination would allow for the evaluation of the scalability of the proposed approach, improve the robustness of the conclusions, and increase the representativeness of the model. The use of more advanced computational infrastructure would also open up the possibility of exploring more complex techniques, optimizing processing times, and further refining the evaluation metrics adopted.